Social Media Monitoring & Alerts System (Week 4 Assessment)

1. Overview

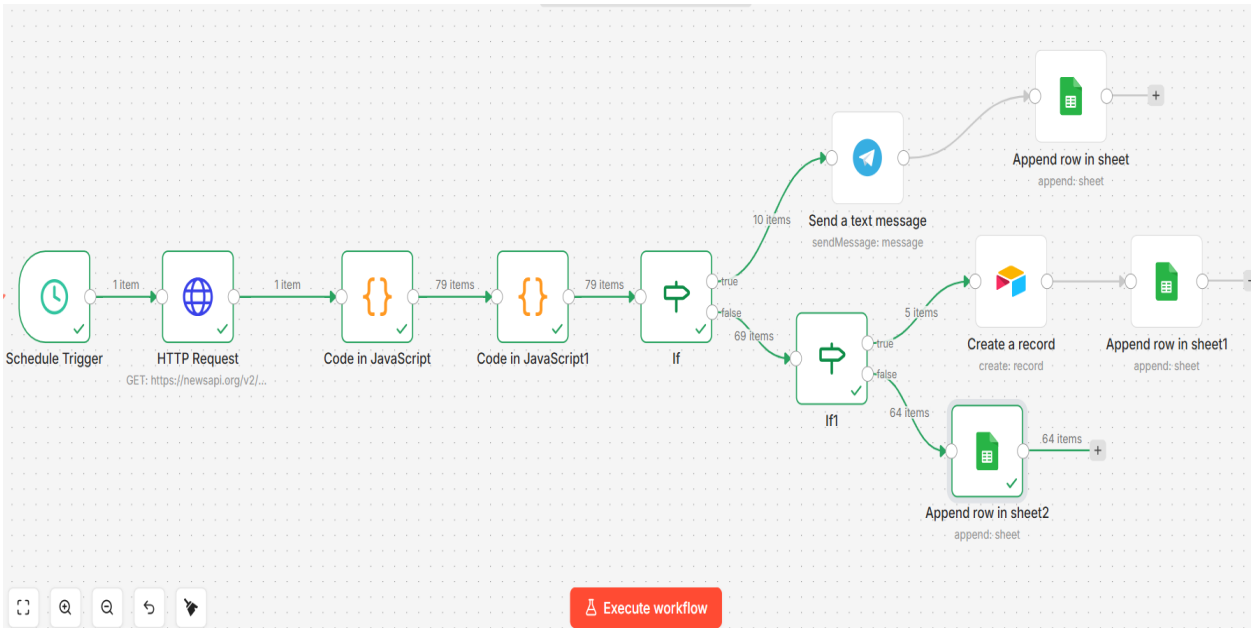
This project implements an automated social media and news monitoring system using n8n. The system tracks online mentions of selected keywords such as “AI”, “cybersecurity”, and “banking”, classifies sentiment using rule-based logic (without AI), and routes the results to appropriate destinations.

The solution is designed to simulate real-world monitoring systems used in cybersecurity and business intelligence, where events are collected, analyzed, and escalated automatically.

2. Objectives

The system achieves the following objectives:

- Automatically fetches articles mentioning specific keywords
- Runs on a scheduled interval (every 15 minutes)
- Classifies sentiment using rule-based keyword matching
- Sends alerts for negative mentions via Telegram
- Stores positive mentions in Airtable for testimonials
- Logs all results (positive, negative, neutral) in Google Sheets



3. Tools and Technologies Used

Tool	Purpose
n8n	Workflow automation and orchestration
NewsAPI	Fetching real-time news articles
Telegram	Sending alerts for negative mentions
Airtable	Storing positive mentions (testimonials)
Google Sheets	Logging all processed data

4. System Architecture

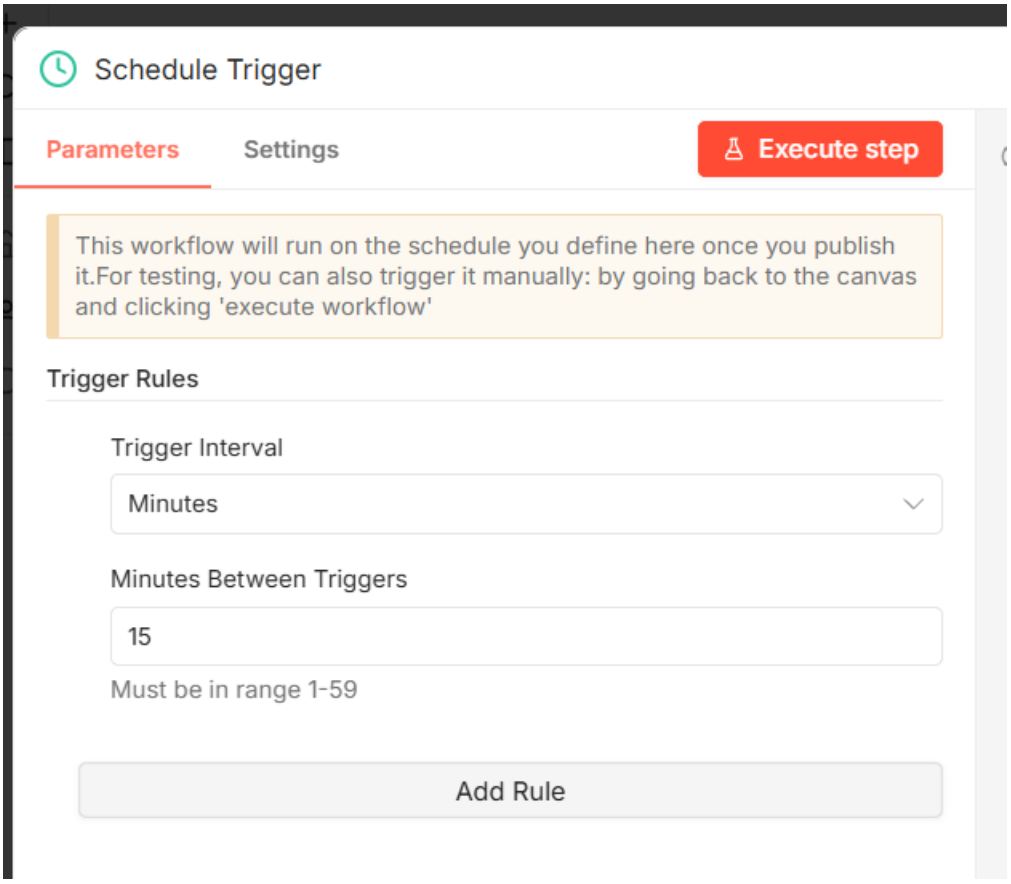
The workflow follows a structured pipeline:

1. Cron Scheduler triggers the workflow every 15 minutes
2. HTTP Request fetches articles from NewsAPI
3. Data Extraction transforms API response into individual items
4. Sentiment Classification applies rule-based keyword matching
5. Routing Logic directs items based on sentiment:
    - Negative → Telegram Alert
    - Positive → Airtable
    - Neutral → No action
6. Google Sheets logs all items for reporting

5. Workflow Implementation (n8n)

5.1 Scheduler (Cron Node)

- Configured to run every 15 minutes
- Ensures continuous monitoring without manual intervention

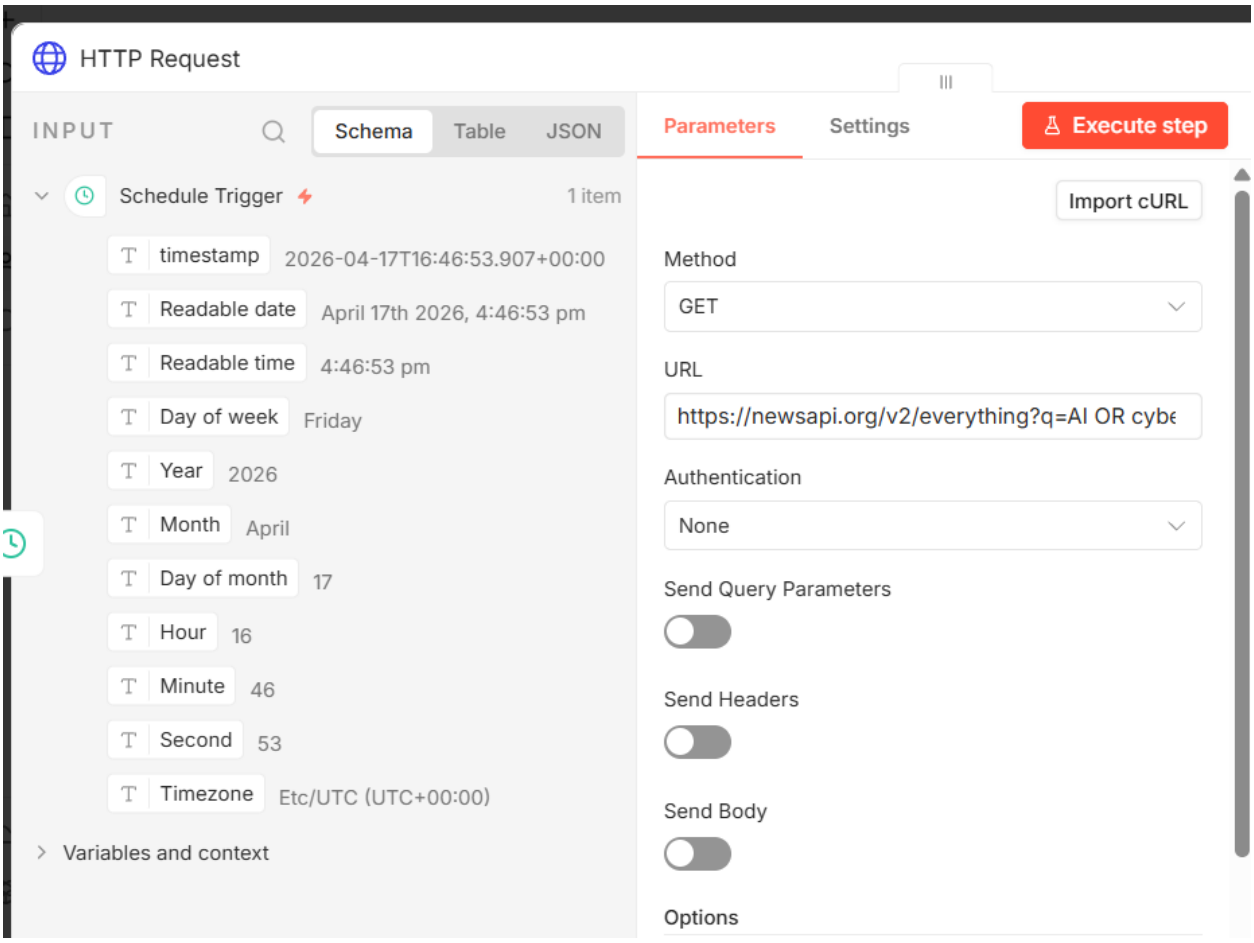


5.2 Data Fetching (HTTP Request Node)

- API used: NewsAPI
- Endpoint: https://newsapi.org/v2/everything?q=AI OR cybersecurity OR banking
- Query parameters include keywords such as: "AI", "cybersecurity", "banking"

This node retrieves articles including:

- Title
- Description
- URL

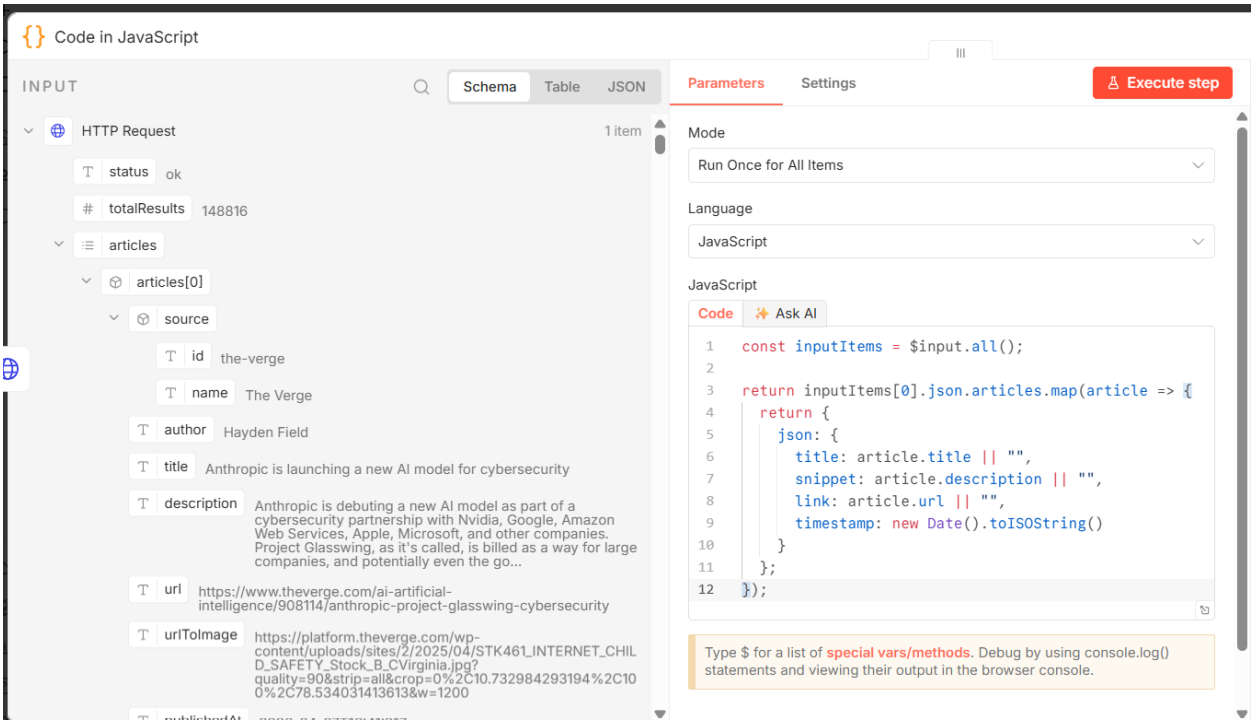


5.3 Data Transformation (Code Node)

The API response contains an array of articles within a single item. A Code node is used to split this into individual items.

Key transformation steps:

- Extract title, description, and URL
- Rename fields for consistency:
  - description → snippet
  - url → link
- Add timestamp for logging



5.4 Sentiment Classification (Code Node)

Sentiment is determined using rule-based keyword matching.

Negative Keywords:

- scam, fraud, bad, issue, broken, hack, breach, fail, leak

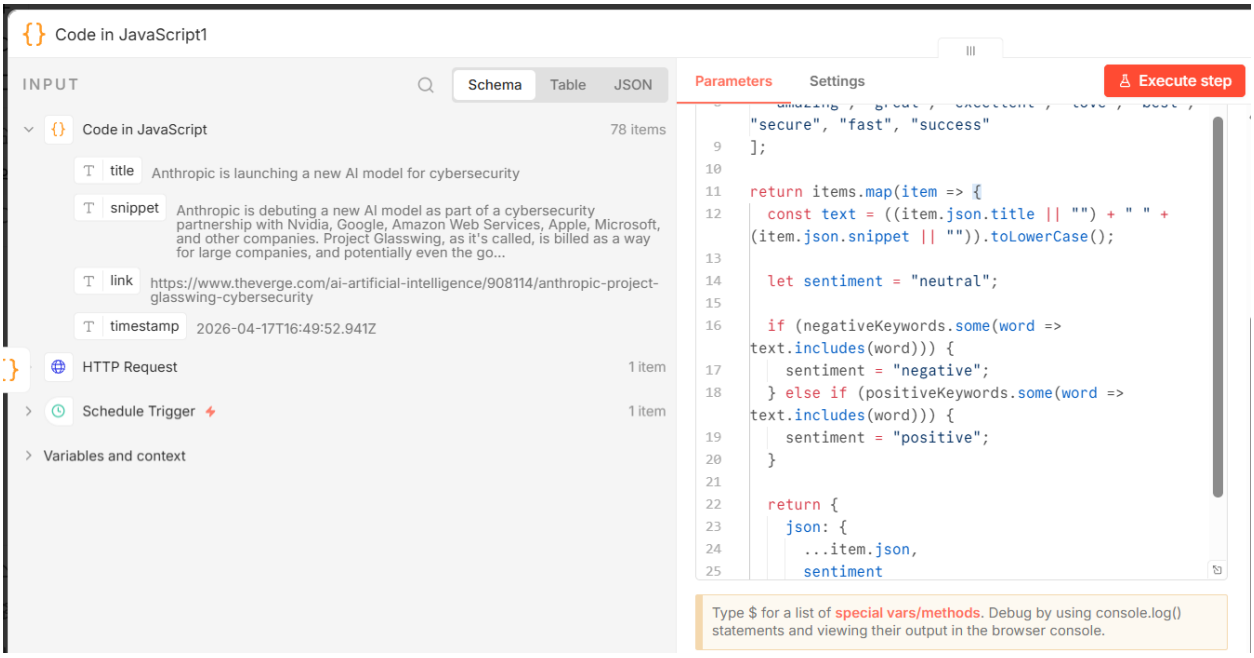
Positive Keywords:

- amazing, great, excellent, love, best, secure, fast, success

Logic:

- If text contains any negative keyword → Negative
- Else if text contains any positive keyword → Positive
- Otherwise → Neutral

This approach avoids AI and ensures transparency and simplicity.



5.5 Routing Logic (IF Nodes)

Two IF nodes are used:

IF Node 1:

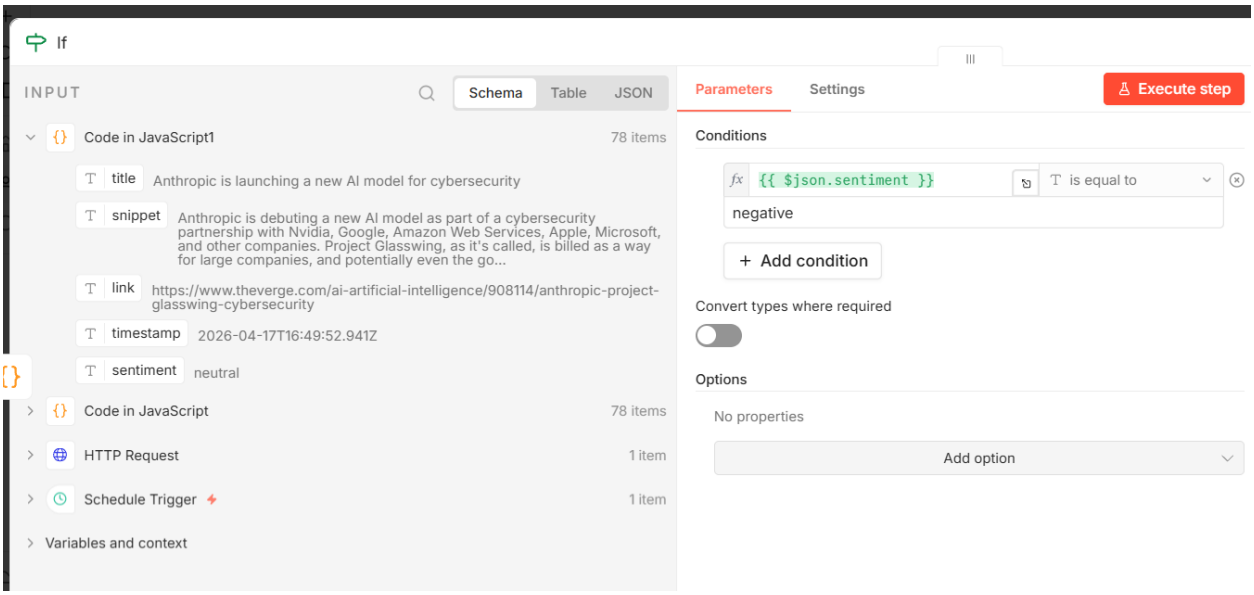
- Condition: sentiment = negative
- Action: Send Telegram alert

IF Node 2:

- Condition: sentiment = positive
- Action: Store in Airtable

Neutral:

- No additional action, only logged

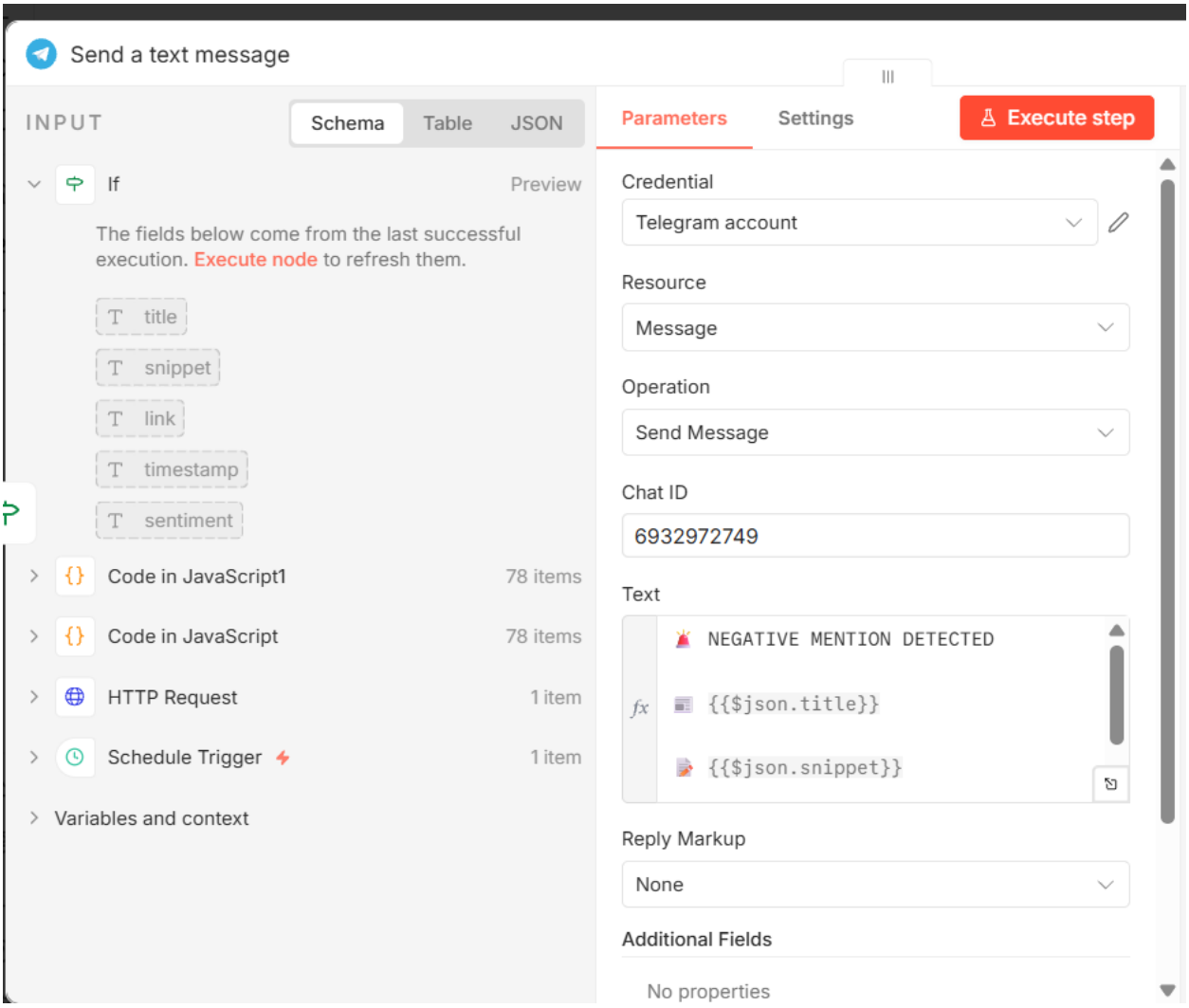


5.6 Alert System (Telegram Integration)

When a negative mention is detected, a message is sent containing:

- Article title
- Description (snippet)
- Source link

This ensures real-time awareness of potential risks.



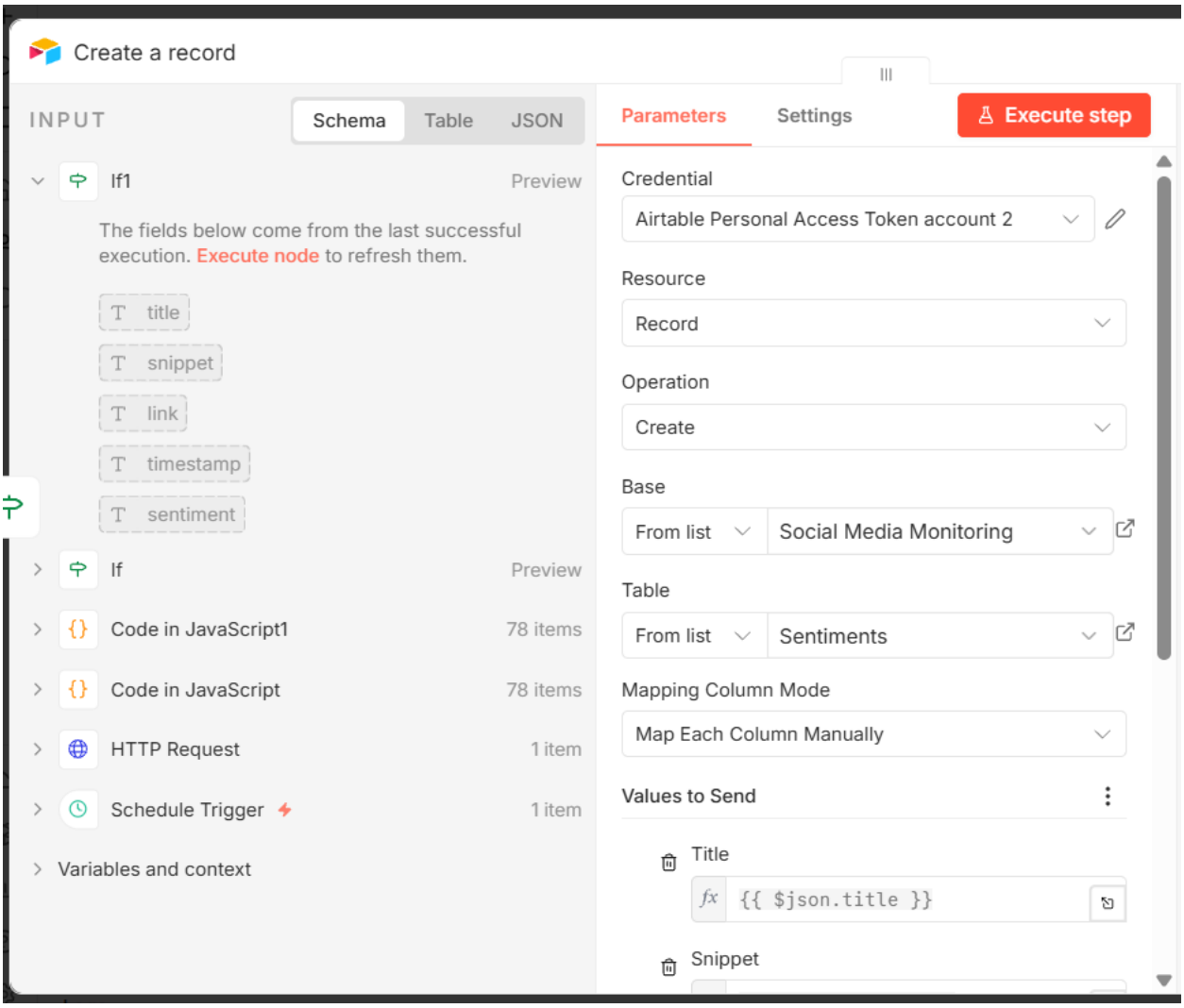
5.7 Positive Mentions Storage (Airtable)

Positive mentions are stored in an Airtable table called “Testimonials”.

Fields stored:

- Title
- Snippet
- Link
- Timestamp

This allows reuse for marketing or reputation management.



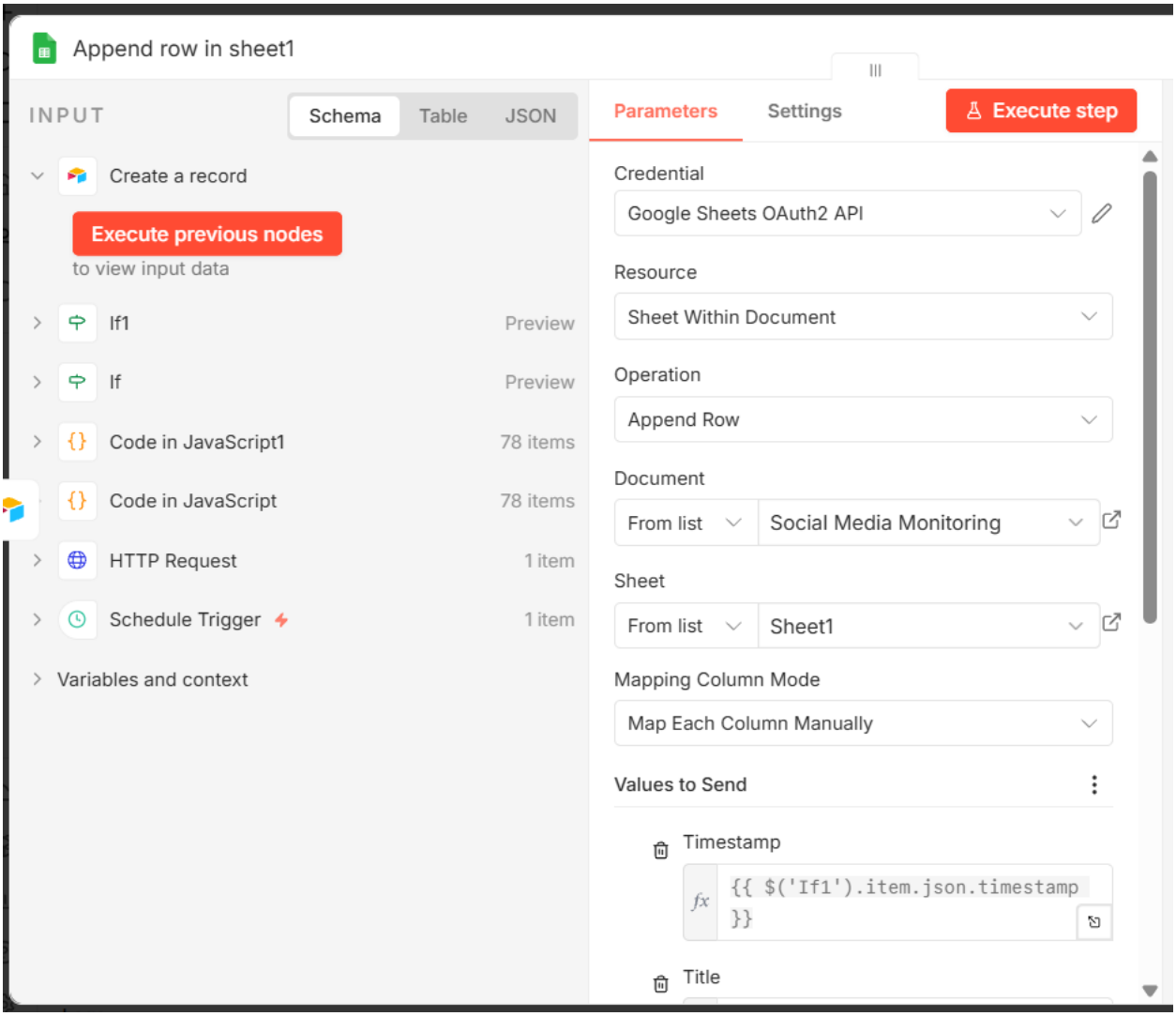
5.8 Logging System (Google Sheets)

All processed items are logged in Google Sheets.

Columns include:

- Timestamp
- Title
- Snippet
- Sentiment
- Link

This provides centralized reporting and historical tracking.



6. Testing and Validation

The system was tested end-to-end using multiple scenarios.

6.1 Scheduler Test

- Workflow triggered automatically every 15 minutes
- Manual execution also verified

6.2 Sentiment Classification Test

Input	Expected Output
“Security breach detected”	Negative

“Amazing AI innovation”	Positive
“New update released”	Neutral

Results matched expected classifications.

6.3 Alert Test

- Telegram alerts triggered only for negative mentions
- No alerts for positive or neutral items

6.4 Airtable Test

- Only positive mentions were stored
- Data correctly mapped to fields

6.5 Google Sheets Test

- All items (positive, negative, neutral) were logged
- No missing records observed

7. Challenges and Solutions

Challenge 1: API Response Structure

- Issue: Data returned as a nested array
- Solution: Used a Code node to split into individual items

Challenge 2: Field Name Differences

- Issue: NewsAPI uses “description” and “url”
- Solution: Mapped fields to “snippet” and “link”

Challenge 3: Null Values

- Issue: Some articles had missing descriptions
- Solution: Used fallback values (empty strings)

8. Future Improvements

The system can be extended in several ways:

- Integrate Twitter/X API for real-time social media monitoring
- Add deduplication to avoid repeated alerts
- Introduce severity levels (e.g., HIGH for “breach”)
- Build a dashboard for visualization
- Replace rule-based logic with NLP/AI models

9. Conclusion

This project demonstrates how automation tools like n8n can be used to build a complete monitoring and alerting system. By combining API integration, rule-based analysis, and multi-channel outputs, the system provides real-time insights into online sentiment.

The solution is scalable, modular, and applicable to real-world use cases such as brand monitoring, threat intelligence, and incident response.